Portfolio Project Final

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MIS380-1: Data Visualization II

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For this Portfolio Project, I have selected the *Employee/HR Dataset* (2023) available on Kaggle. I chose this dataset as it contains a large and varied collection of human resources (HR) information that is organized into several related tables. This structure allows for more comprehensive analysis than a single table dataset and supports the development of a relational data model in Power BI.

Employee/HR Dataset

The dataset contains four main tables: employee_data, employee_engagement_survey _data, recruitment_data, and training_and_development_data. The employee_data contains demographic information, job titles/departments, performance, and other foundational details for the workforce. The employee_engagement_survey_data tables include engagement, satisfaction, and work-life balance scores which offer insights on workplace experience and organizational health. The recruitment_data presents information about candidate location, education levels, years of experience, desired compensation, and candidate requisition status. Lastly, the training_and_development_data provides information on training programs, outcomes, duration, and cost. These tables are connected through shared identifiers such as employee, which makes it possible to build an effective relational model that integrates information across the employee lifecycle.

Prior to visualization and analysis, the dataset will undergo several data cleaning and transformation steps to ensure accuracy and readiness within Power BI. These steps include reviewing each table for missing or null values, standardizing column names and data types, and verifying the consistency of shared identifiers used to relate tables (such as employee IDs). Where appropriate, categorical values such as job titles, departments, or recruitment statuses may be normalized to reduce duplication and improve filtering. Date fields related to hiring, training, or survey responses will be formatted consistently to support time-based analysis. Additionally, unnecessary or redundant columns will be removed to streamline the data model and improve performance. These preparation steps help ensure the dataset is reliable, analytically sound, and well-structured for building meaningful measures, relationships, and visualizations in Power BI.

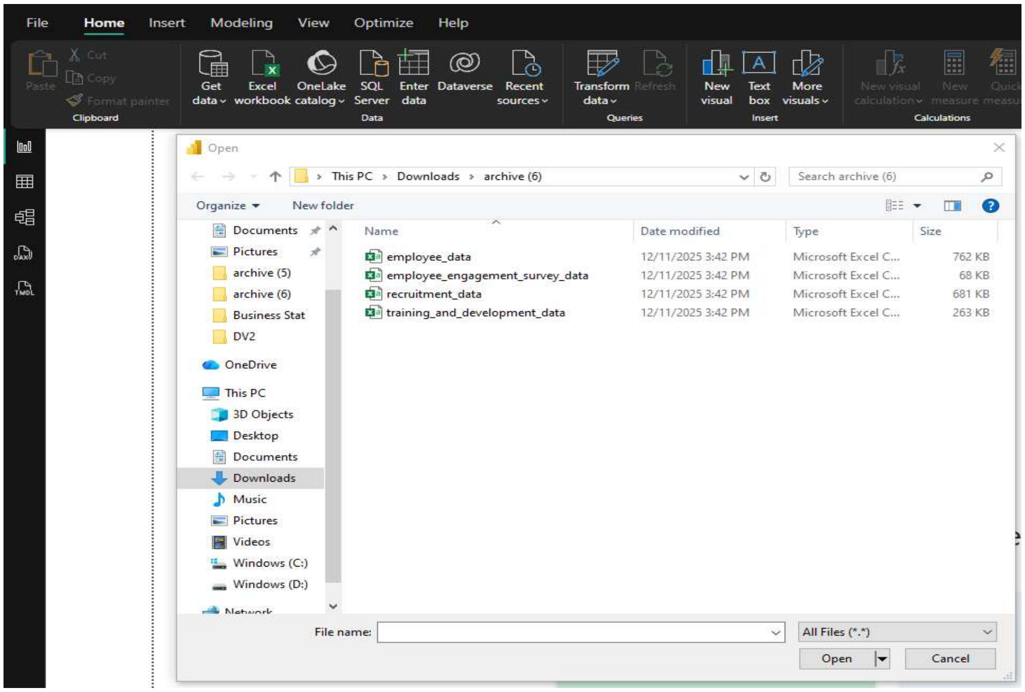
This dataset was appealing because its multiple tables provide a broad view of workforce activity that can be examined from several perspectives. The different files connect information about employees, engagement levels, hiring outcomes, and development programs allowing for deeper exploration of how these elements are related to one another within the organization. The structure of the dataset makes it possible to study patterns such as engagement scores vary across job categories, how development activities may influence overall employee satisfaction, or how demographic characteristics connect to workplace trends. The ability to integrate these tables creates a more realistic representation of organizational dynamics and supports more thoughtful analysis of HR outcomes. My background within this field gives me practical experience in understanding how these types of metrics are used in decision making. It supports my ability to interpret findings and develop actionable insights that reflect real organizational challenges and opportunities.

Employee/HR Transformation

The Employee/HR Dataset was imported into Power BI using individual source files rather than a combined dataset. Figures 1 & 2 show each file was uploaded one at a time using the Get Data feature in Power BI and selecting the CSV file format. Importing the tables individually allowed greater control during the transformation process, as each dataset required different cleaning steps, data type adjustments, and validation checks prior to modeling. This approach also supported clearer identification of each table's purpose and structure before relationships were established within the data model.

Figure 1

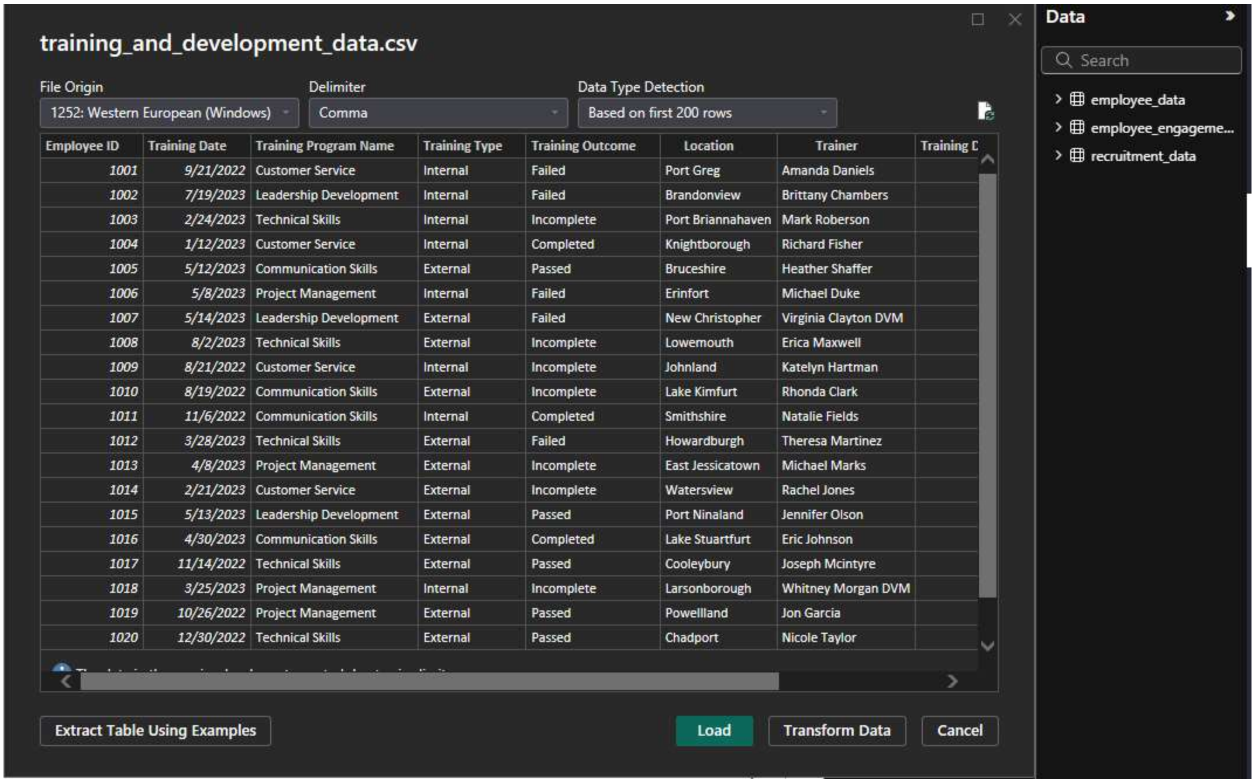
Importing Employee/HR Dataset CSV Files into Power BI



Note. This figure illustrates the process of importing the Employee/HR Dataset into Power BI by selecting individual CSV files using the Get Data feature. The dataset includes separate files for employee information, engagement surveys, recruitment data, and training and development records sourced from Kaggle (Rana, 2023).

Figure 2

Preview of Data Imported into Power BI



Employee ID	Training Date	Training Program Name	Training Type	Training Outcome	Location	Trainer	Training D
1001	9/21/2022	Customer Service	Internal	Failed	Port Greg	Amanda Daniels	
1002	7/19/2023	Leadership Development	Internal	Failed	Brandonview	Brittany Chambers	
1003	2/24/2023	Technical Skills	Internal	Incomplete	Port Briannahaven	Mark Roberson	
1004	1/12/2023	Customer Service	Internal	Completed	Knightborough	Richard Fisher	
1005	5/12/2023	Communication Skills	External	Passed	Bruceshire	Heather Shaffer	
1006	5/8/2023	Project Management	Internal	Failed	Erinfort	Michael Duke	
1007	5/14/2023	Leadership Development	External	Failed	New Christopher	Virginia Clayton DVM	
1008	8/2/2023	Technical Skills	External	Incomplete	Lowemouth	Erica Maxwell	
1009	8/21/2022	Customer Service	Internal	Incomplete	Johnland	Katelyn Hartman	
1010	8/19/2022	Communication Skills	External	Incomplete	Lake Kimfurt	Rhonda Clark	
1011	11/6/2022	Communication Skills	Internal	Completed	Smithshire	Natalie Fields	
1012	3/28/2023	Technical Skills	External	Failed	Howardburgh	Theresa Martinez	
1013	4/8/2023	Project Management	External	Incomplete	East Jessicatown	Michael Marks	
1014	2/21/2023	Customer Service	External	Incomplete	Watersview	Rachel Jones	
1015	5/13/2023	Leadership Development	External	Passed	Port Ninaland	Jennifer Olson	
1016	4/30/2023	Communication Skills	External	Completed	Lake Stuartfurt	Eric Johnson	
1017	11/14/2022	Technical Skills	External	Passed	Cooleybury	Joseph Mcintyre	
1018	3/25/2023	Project Management	Internal	Incomplete	Larsonborough	Whitney Morgan DVM	
1019	10/26/2022	Project Management	External	Passed	Powellland	Jon Garcia	
1020	12/30/2022	Technical Skills	External	Passed	Chadport	Nicole Taylor	

Note. This figure displays the preview of the CSV files during import into Power BI.

Transforming the Data Tables

After importing the dataset into Power BI, the next step involved transforming and cleaning the data to ensure it was suitable for accurate analysis and modeling. Data transformation is a critical phase in the analytics process, as it improves data quality, consistency, and usability. As Geerts (2023) explains, “Data transformation refers to profiling, cleaning, filtering, and integrating data, all of which prepares the data for analysis purposes” (para. 1). Using Power Query Editor, each table was reviewed individually to address data type inconsistencies, standardize naming conventions, resolve duplicate records where appropriate, and improve overall readability. These transformations ensured the dataset was structured in a way that supports reliable analysis and effective relationship modeling within Power BI.

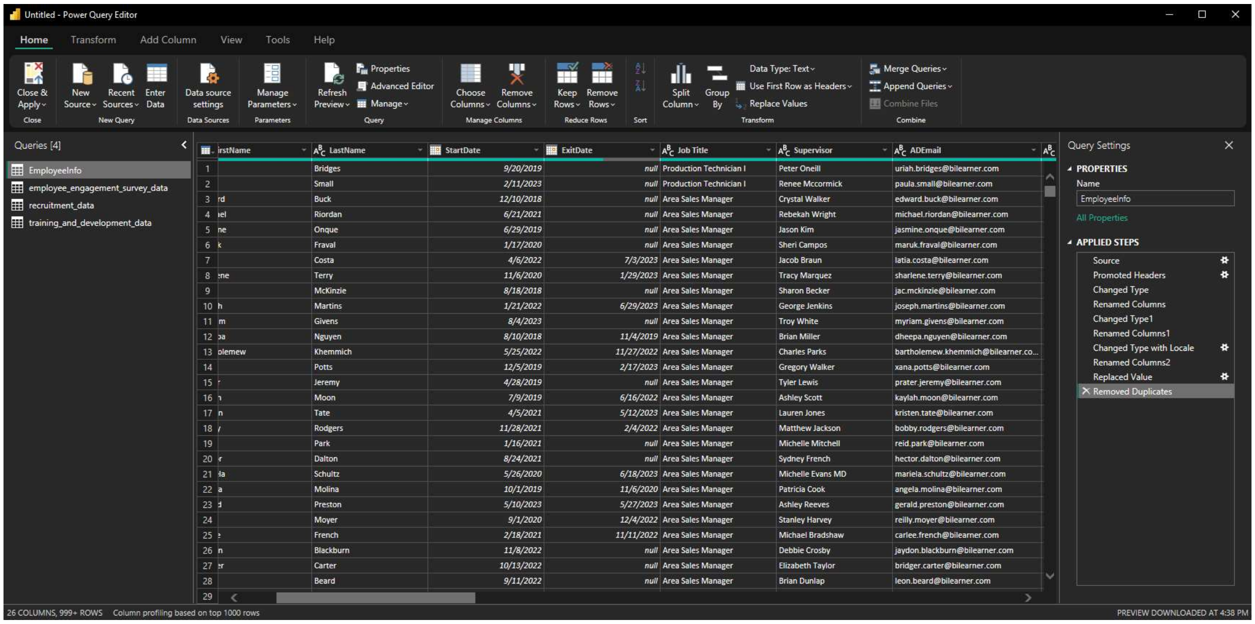
In addition to standardizing data types and column naming, each table was reviewed for missing or null values during the transformation process. This review focused on key identifier fields, date fields, and numeric measures to ensure they would not interfere with relationship integrity or analytical calculations. While some optional or descriptive fields contained null values, no critical issues were identified that would prevent accurate modeling or visualization. Where null values appeared in non-essential fields, they were retained to preserve the original structure of the dataset rather than introducing assumptions through imputation. This validation step confirmed that the dataset was sufficiently complete and reliable for use in Power BI without requiring extensive corrective handling.

Employee_Data Table

As shown in Figure 3, several transformations were applied to the Employee data table to improve consistency, accuracy, and suitability for relational modeling. The table was first renamed to Employee to provide a clear and concise identifier within the data model. The EmpID column was renamed to EmployeeID for consistency with other tables and converted from numeric to text to ensure it is treated strictly as an identifier rather than a quantitative value, preserving data integrity and preventing unintended numeric operations. Additional column standardization included renaming the Title field to JobTitle for clarity and readability. The Date of Birth (DOB) column was converted from text to a date data type using a United Kingdom locale to ensure correct day and month interpretation and to align its formatting with start and exit date fields. Text values within the TerminationType column were also standardized by replacing abbreviated entries such as “Unk” with “Unknown” to improve interpretability. Finally, duplicate records were removed based on the Employee ID column to ensure each employee was represented only once, establishing the table as a reliable primary dimension for subsequent relationships.

Figure 3

Employee_Data Table Transformation



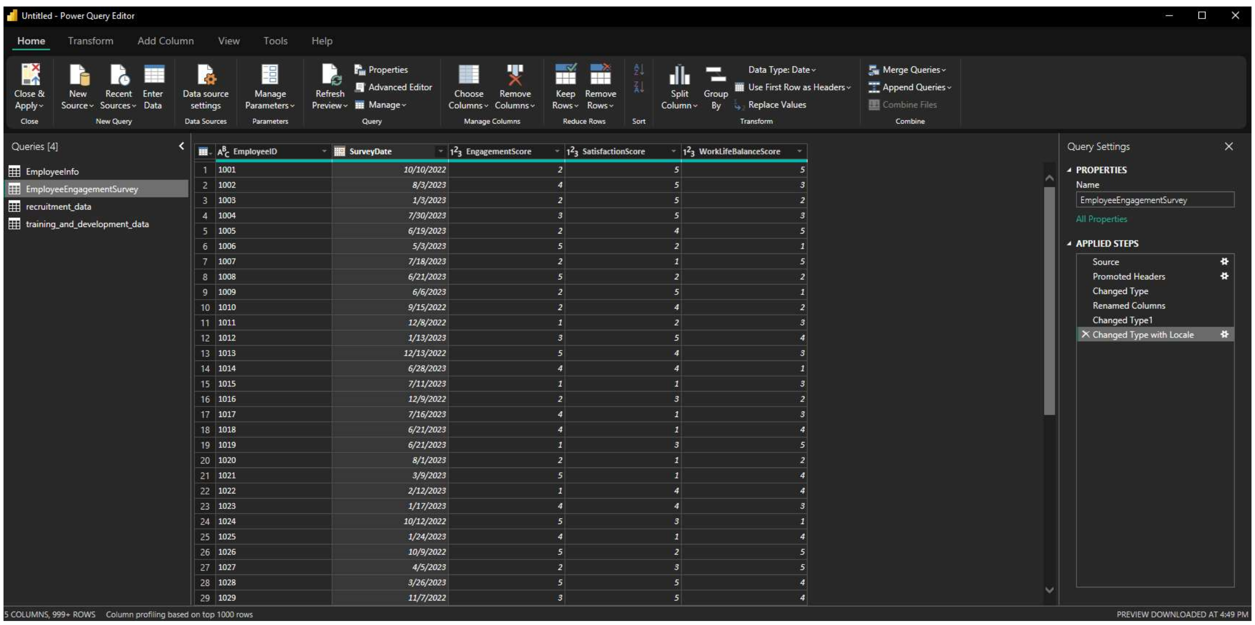
Note. This figure shows the Employee data table after transformations were applied in Power Query Editor, including column renaming, data type standardization, text cleanup, and duplicate record removal to prepare the table for use as a primary employee reference table.

Employee_Engagement_Survey_Data Table

As illustrated in Figure 4, targeted transformations were applied to the Employee Engagement Survey data to ensure consistency with the overall data model while preserving the integrity of survey responses. The table was renamed to EngagementSurvey to provide a concise and descriptive identifier within the model. The EmployeeID column was converted from numeric to text to align with its role as a foreign key and to ensure compatibility with the Employee table. Column names were reformatted using camel case conventions, including standardizing Employee ID to EmployeeID, to maintain consistent naming practices across all tables. The SurveyDate field was converted from text to a date data type using the same locale settings applied to other date fields, ensuring accurate interpretation and uniform formatting. Duplicate records were intentionally not removed from this table, as employees may complete multiple surveys over time. Each row represents a distinct survey response, and removing duplicate EmployeeIDs would have eliminated valid historical observations necessary for analyzing trends in employee engagement.

Figure 4

Employee_Engagement_Survey_Data Table Transformation



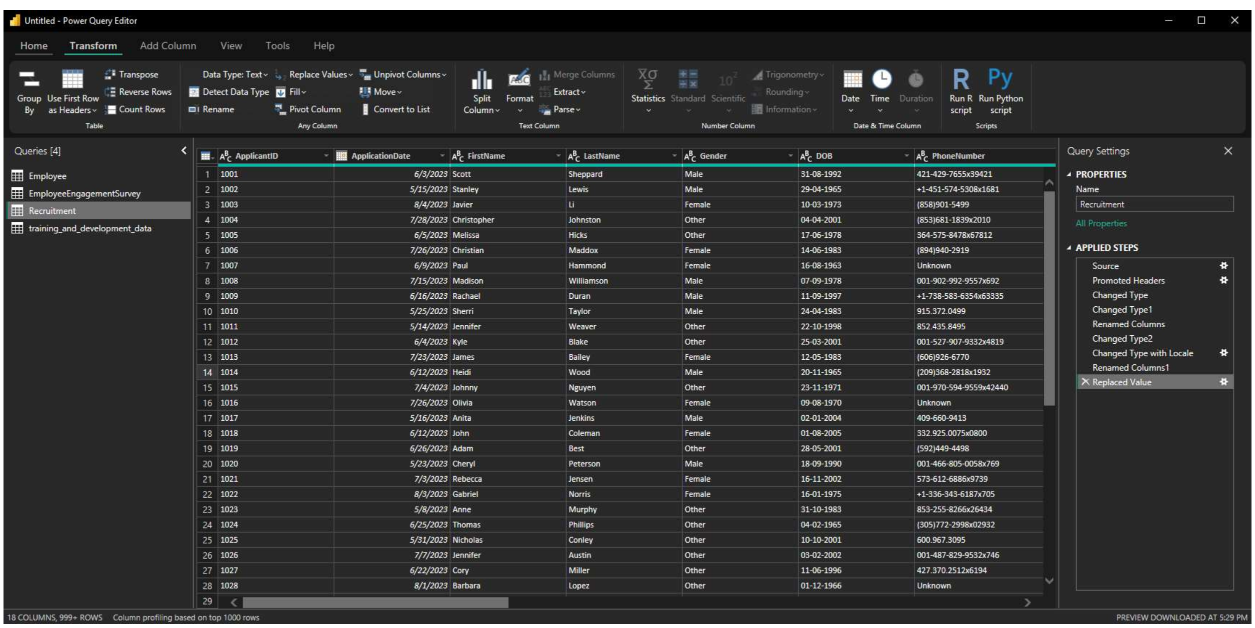
Note. This figure displays the EngagementSurvey table after data type standardization and column renaming were applied in Power Query Editor. The table contains survey response records, including engagement, satisfaction, and work-life balance scores, which were retained at the individual response level to support longitudinal analysis.

Recruitment_Data Table

As shown in Figure 5, several transformations were applied to the Recruitment data table to improve consistency and clarity while preserving the integrity of applicant-level information. The table was renamed to Recruitment to align with the established naming conventions used throughout the model. The ApplicationID column was converted from numeric to text to ensure it functioned strictly as an identifier rather than a quantitative value, and column names were reformatted using camel case for consistency, including standardizing Application ID to ApplicationID. The Date of Birth field was renamed to DOB and converted from text to a date data type using the same locale settings applied across other tables to ensure accurate interpretation and formatting. The Desired Salary column was converted to a fixed decimal number using a United States locale to support consistent numeric representation. Phone number values were retained as text due to global formatting differences, the presence of extensions, and placeholder entries. Obvious placeholder values such as “####” were replaced with “Unknown” for clarity, and the field will not be used for analytical calculations. Additionally, address data revealed inconsistencies between country, state, and postal code fields, indicating unreliable geographic information. To avoid introducing inaccuracies, geographic values were retained as provided and will either be analyzed independently or will be excluded from combined geographic analysis.

Figure 5

Recruitment_Data Table Transformation



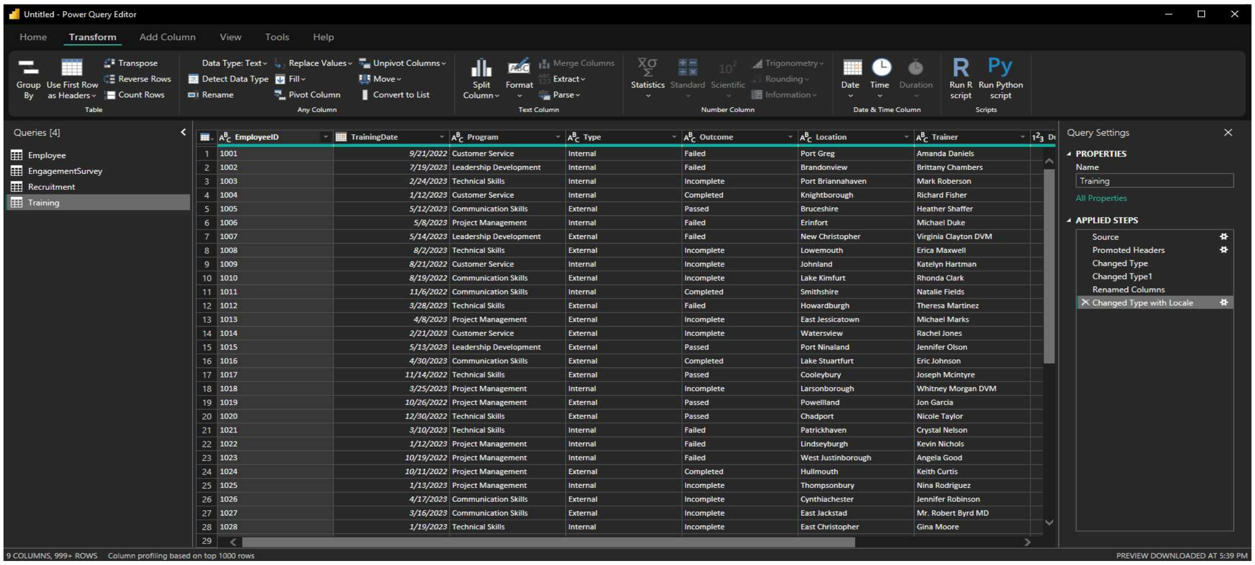
Note. This figure displays the Recruitment table after transformations were applied in Power Query Editor, including data type standardization, column renaming, text cleanup, and selective handling of inconsistent contact and geographic information to preserve data accuracy.

Training_and_Development_Data Table

As illustrated in Figure 6, several transformations were applied to the training and development data to ensure consistency with the overall data model and to support accurate analysis of employee learning activities. The table was renamed to Training to align with the established naming conventions used across the dataset. The EmployeeID column was converted from numeric to text to ensure it functioned strictly as an identifier and could reliably serve as a foreign key when establishing relationships with the Employee table. Column names were reformatted using camel case conventions for consistency, and repetitive wording was removed from several fields to improve readability, such as simplifying TrainingProgram to Program and TrainingType to Type. The Cost column was converted to a fixed decimal number using a United States locale to ensure consistent currency representation. Duplicate records were intentionally not removed from this table, as employees may complete multiple training courses over time. Each row represents a distinct training event, and removing duplicate EmployeeIDs would have eliminated valid historical observations necessary for analyzing training participation and outcomes.

Figure 6

Training_and_Development Table Transformation



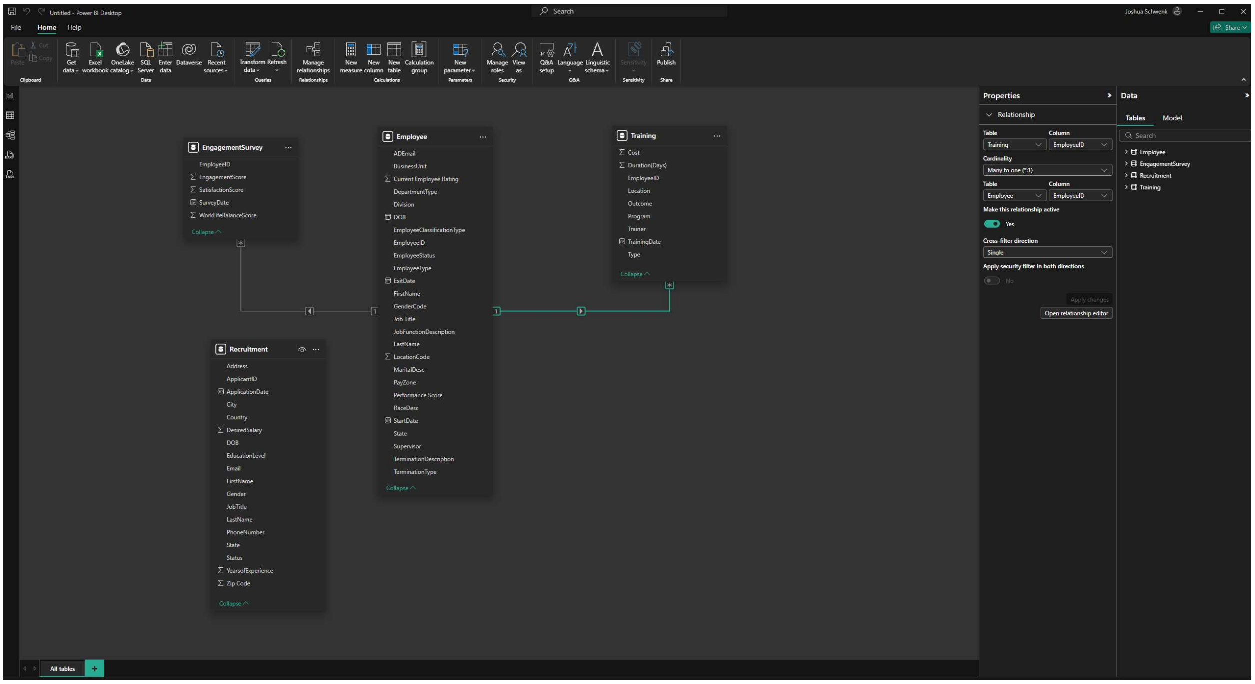
Note. This figure displays the Training table after transformations were applied in Power Query Editor, including column renaming, data type standardization, and formatting adjustments to support analysis of employee training participation and outcomes.

Data Model

The data model was designed using a dimensional structure that supports clear, accurate analysis while minimizing ambiguity. In Power BI, the most commonly used relationship type is the one-to-many structure, which represents scenarios where a single record in a dimension table relates to multiple records in fact-style tables. In this model, the Employee table serves as the central dimension, with each employee represented by a single unique EmployeeID. This table is connected to both the EngagementSurvey and Training tables through many-to-one relationships, as individual employees may complete multiple engagement surveys and participate in multiple training activities over time. These relationships are implemented with single-direction filtering, allowing the Employee table to control how related fact records are filtered, which helps maintain consistent and interpretable results. While Power BI supports other relationship types, such as one-to-one and many-to-many, these were not required for this dataset. Many-to-many relationships were avoided due to their potential to introduce ambiguity and duplicate counting unless carefully resolved through additional bridge tables (Mannino, 2022). The Recruitment table was intentionally left unconnected, as it represents applicants rather than employees and does not contain a stable foreign key that aligns with the Employee table. Fields such as names, email addresses, or dates of birth were not considered reliable enough to establish a true relationship, and forcing a connection could have introduced inaccuracies into the model.

Although the Recruitment table remains intentionally disconnected in the current model, a future iteration could incorporate a bridge or staging table if a reliable recruitment-to-employee linkage were introduced. For example, a staging table could be used to map applicants to hired employees once a confirmed EmployeeID or system-generated hire identifier becomes available. This approach would allow recruitment metrics, such as source effectiveness or time-to-hire, to be analyzed alongside post-hire engagement and training outcomes while avoiding the risks associated with many-to-many or inferred relationships. At present, this structure is not implemented to preserve model accuracy, but the design allows for scalable enhancement should the dataset evolve.

Figure 7
Employee-Centered Data Model with Fact and Dimension Tables



Note. This figure illustrates the final data model implemented in Power BI, showing the Employee table as the central dimension with one-to-many relationships to the EngagementSurvey and Training tables. The Recruitment table is intentionally left unconnected due to the absence of a reliable shared identifier.

References

Geerts, G. (2023). *Introduction to Microsoft Power BI.* Zybooks.

Mannino, M. (2022). *Database design: Design, query, formulation, and administration using Oracle and PostgreSQL*. SAGE Publications. ISBN: 978-1948426954

Rana, R. S. (2023). *Employee/HR Dataset (All in One)* [Dataset]. Kaggle. https://www.kaggle.com/datasets/ravindrasinghrana/employeedataset